What should my store sell?

BIG DATA ENGINEERING

Using data to decide what my store should sell

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Can this scale?

A reusable process for product decisions

01 MARKET SEARCH



LEGO Botanicals

Launched 2021 / standalone theme 2025

Bestselling theme: H1 2025 & H1 2026 [1]

Other flower toys?

Explore new categories

Still brick flowers? ✓

Find the unmet needs

02 BIG DATA PROJECT

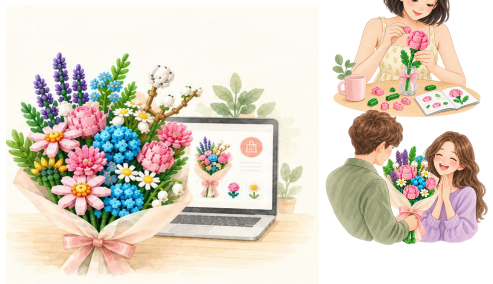


Demand beyond LEGO.
Room to improve.

80%
Non-LEGO review share

0.57
points below LEGO [2]

03 MY STORE



Half the price.
More choice.

Target: half the price of comparable LEGO sets

1 pipeline × N categories

Toys today.
Other categories next.

[1] LEGO H1 2025 & H1 2026 results; theme announcement (2024). [2] Amazon Reviews 2023 analysis. Reviews are not sales.

Overview: 2023 Finds, 2026 Verifies

One pipeline, used twice: 2023 guides product selection, 2026 supports the choice, and my store uses the evidence.



1 PIPELINE

① Data sources

- **FIND · BUILD**
  US Amazon 2023
  16.26M reviews
  free public dataset
- **VERIFY · REUSE**
  US Amazon 2026
  724 product families
  paid API

→

dagster Orchestration

② Ingestion — Python + DuckDB
read the files
10 GB, processed locally
no server

→

③ Loading — Meltano
load to the cloud

→

④ Storing — BigQuery
raw data, kept as loaded

→

⑤ Transform — dbt
four layers, star schema

→

⑥ Testing — dbt + Great Expectations
dbt tests + GE
106 dbt tests, 24 GE rules

2 ANALYSE

Streamlit app
explore the evidence

Notebook
SQLAlchemy + pandas

→

3 ANSWER

2023 FINDS

2.76×

relative growth index vs all toys

2026 VERIFIES

+194%

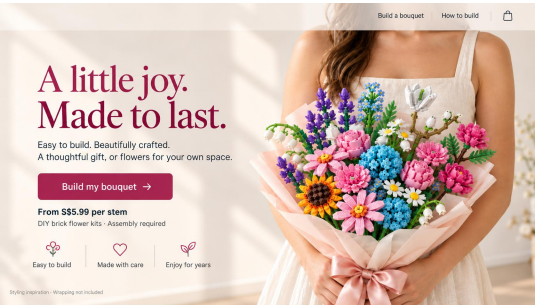
ratings added vs 2025

→

4 OUTPUT

My store
what to sell

Sell brick flowers.



Source: FIND — Amazon Reviews 2023 (McAuley Lab, UCSD) ;VERIFY — Keepa API, Amazon.com (accessed Oct 2026)

Ingest, Load and Store: Raw Files to BigQuery

Two raw files, 890,874 products and 16.26M reviews, land in BigQuery unchanged and checked.



1 INGEST read the files

PRODUCTS `meta_Toys_and_Games.jsonl.gz`

REVIEWS `Toys_and_Games.jsonl.gz`

DuckDB on my laptop: 10 GB, no server

890,874 products

16.26M reviews

CHECK **0 null keys; 0 duplicate product keys**

2 LOAD move them to the cloud

DuckDB one row — EXAMPLE: LEGO Orchid

parent_asin	title	price
B09Q4L157D	LEGO Icons Orchid…	40.0

meltano

Singer: the tap sends each row as a JSON message
the target loads it into BigQuery

BigQuery the same row, one JSON cell

```
{"parent_asin": "B09Q4L157D",
 "title": "LEGO Icons…", "price": 40.0}
```

Same values, new shape. Dbt unpacks it next.

CHECK **890,874** rows in = **890,874** rows out

3 STORE keep them as the raw layer

BigQuery four datasets, one per layer

project2_analytics	BigQuery	Default	US
project2_marts	BigQuery	Default	US
project2_raw	BigQuery	Default	US
project2_staging	BigQuery	Default	US

This page fills **project2_raw**. dbt builds the rest:

4 **MARTS** the answers

3 **ANALYTICS** the star schema

2 **STAGING** typed columns

1 **RAW** we are here

RULE **Never edit Raw.** Fix it higher up.

Transform (1/2): Four Layers, One Review

Dbt builds three layers on top of Raw. One real review shows what changes at every step.



LAYER	THE SAME REVIEW (EXAMPLE: one review of the LEGO Orchid)	WHAT CHANGES	BUILT FROM
1 **RAW** Keep the original	`{"parent_asin": "B09Q4L157D", "rating": 5, "timestamp": 1675286757274, ...}`	**Start** one JSON cell, exactly as loaded	**Meltano load**
2 **STAGING** Make it readable	parent_asin: B09Q4L157D; rating: 5.0; review_datetime: 2023-02-01 21:25	**Unpack** typed columns; the number is now a time	**Layer 1**
3 **ANALYTICS** Build the star schema Add flower-toy type	fact_reviews (1 row = 1 review): parent_asin FK: B09Q4L157D; rating: 5.0; review_datetime: 2023-02-01 21:25 dim_flower_product ↓ FK → PK (1 row = 1 flower toy): parent_asin PK: B09Q4L157D; title: LEGO Orchid; play_pattern DERIVED: Brick flower bouquets	**Link** The product foreign key links each review to its product.	**Layer 2**
4 **MARTS** Answer the question: which kind of flower toy is growing?	play_pattern: Brick flower bouquets; reviews 2023: 2,002; avg rating: 4.29; Index vs all toys: 2.76×	**Performance** reviews grouped by play pattern relative growth compared with all toys	**Layer 3** more on the next page

How to find flower toys and their types:

1 Search flower keywords in titles
890,874 → 37,441
2 Keep relevant flower toys → 3,650
3 Label by use → 22 play patterns
e.g. brick bouquets, planting kits, plush, puzzles

Transform (2/2): The Star Schema and the Marts

Layer 3 is a star: reviews in the middle, the tables that describe them around. Layer 4 turns it into the answer.



1 STAR SCHEMA Layer 3 Structure

dbt

1 row of the fact = 1 review.
Its keys point to the tables that describe it.

WHEN?

dim_date 8,352
- PK **date_key**
- year, month, day

WHO?

dim_customer 8,116,226
- PK **user_id**
- reviews written

dim_flower_product 3,650
- PK **parent_asin**
- play_pattern, brand

WHICH KIND? from the flower labels

fact_reviews 16,260,406
- FK **date_key**
- FK **user_id**
- FK **parent_asin**
- rating
- review_datetime

WHAT PRODUCT?

dim_product 890,874
- PK **parent_asin**
- title, price

1 product **many** pairs

bridge many-to-many 2.9M
- FK **parent_asin**
- FK **category_key**

many pairs **1** category

dim_category 924
- PK **category_key**
- category name

Why a bridge? The Orchid: 2 categories, 522 reviews. MANY-TO-MANY

THE TABLES / THE RESULT

WITHOUT a bridge

product	category
Orchid	Toys & Games
Orchid	Most-Loved Toys

2 rows →

product	reviews
Orchid	1,044

wrong

WITH a bridge

product
Orchid

1 row +

product	category
Orchid	Toys & Games
Orchid	Most-Loved Toys

two pairs moved their own table →

product	reviews
Orchid	522

right

2 THE CORE MART Layer 4 Subject-specific data

QUESTION
Which kind of flower toy is growing?

1. **Link** each review to its play pattern
2. **Count** reviews in 2022 and 2023
3. **Compare** growth with all toys

play pattern	2022	2023	change	vs market
Brick flower bouquets	**1,135**	**2,002**	**1.76×**	**2.76×**
Toddler garden sets	1,235	795	0.64×	**1.01×**
Craft bouquets	305	174	0.57×	**0.89×**
Flower-image puzzles	1,803	892	0.49×	**0.77×**
Planting kits	149	57	0.38×	**0.60×**
All toys (the market)	*1.42M*	*0.90M*	*0.64×*	*1.00×*

vs market = change ÷ 0.64 1.00 = same as the market

ANSWER

Brick flowers: +76% reviews
Relative growth index vs all toys: 2.76×

Test: A Check at Every Step

Amazon 2023: four checks cover ingestion, loading and outputs. dbt checks model integrity; Great Expectations checks mart results.



FOUR CHECKS ALONG THE PIPELINE each checks a different part

AT INGEST DuckDB
The keys of the 2 raw files
0 empty, 0 duplicate product keys
shown on page 4

→

AT LOAD Meltano
Rows before Meltano = rows after
890,874 in DuckDB = 890,874 in BigQuery
shown on page 4

→

AT TRANSFORM dbt tests
Keys, values and row counts
87 tests
this page

→

AT THE END Great Expectations
Mart outputs: the 3 tables of the answer
15 rules
this page

1 FIRST: DBT TESTS

dbt

DATA MODEL INTEGRITY is every table built right?

	KIND OF CHECK	WHAT IT CHECKS	NUMBER OF TESTS
1	**Null values** `not_null`	No empty key. every review has a product, a date and a customer	34
2	**Duplicates rows** `unique`	Unique keys at each table's defined grain. one row per product, per date, per customer	16
3	**Referential integrity** `relationships`	Every review points to a product that exists. each key in the fact table is found in its dimension	11
4	**Business logic** `value rules + my own SQL`	Value rules and row counts ratings: 1–5; positive prices; row counts preserved	19
5	**Known source issues warn, do not fail**	**Source issues are kept and flagged.** 174,053 duplicate reviews: index 2.76× with them, 2.77× without.	7

87 tests: **80** pass **7** warn on purpose **0 errors**

2 THEN: GREAT EXPECTATIONS

Great Expectations

DATA QUALITY does the answer make sense?

	KIND OF CHECK	WHAT IT CHECKS	NUMBER OF RULES
1	**Complete** `right number of rows`	All 22 kinds of flower toy have a row. nothing got lost on the way	2
2	**Possible** `value in a real range`	Each kind's average rating is 1 to 5. dbt checked each review; this checks my calculation	7
3	**Allowed** `label from a fixed list`	Each evidence tier is strong, medium or weak. no other label, no typing mistake	3
4	**Consistent** `numbers agree`	Reviews about one topic never exceed all reviews. nothing is counted twice	3

15 rules: **15** pass

AMAZON 2023 DATA · FIND

2026 DATA · VERIFY

Analyse: What to Sell, and How to Sell It Better

The numbers pick the product. The reviews show what goes wrong. Together they give my plan.

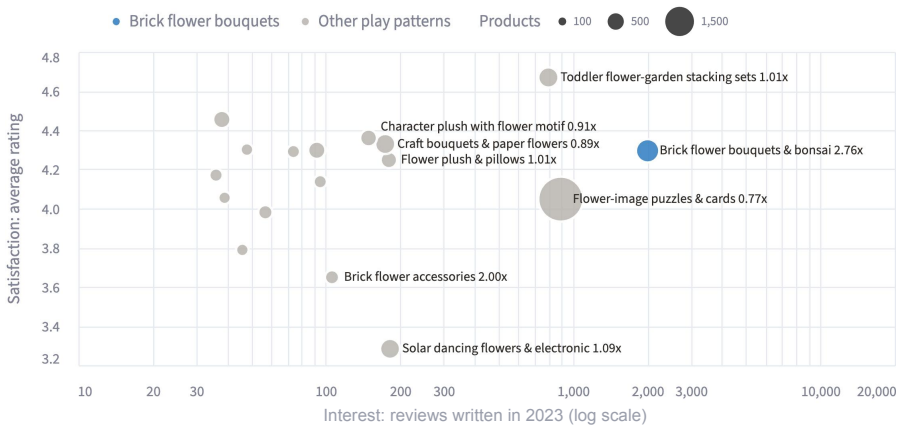


1 WHICH KIND?

Streamlit app

Brick flowers lead in review activity.
Demand extends beyond LEGO.
Other brands received 80% of sampled 2023 brick-flower reviews.

Figure 1. Interest and satisfaction by play pattern
(22 kinds of flower toy, Amazon 2023)



One dot = one kind of flower toy. Right = more reviews in 2023 (Jan-Sep). Up = higher rating. Size = number of products.

WHO GETS THE GROWTH?

Table 1. Brick flower bouquets reviews: LEGO vs other brands

	reviews 2022 → 2023	change	share of reviews	rating
LEGO	427 → 405	**-5%**	**38% → 20%**	4.72
Other brands	**708 → 1,597**	**+126%**	**62% → 80%**	**4.15**

2 WHAT GOES WRONG

Reviews point to improvement opportunities.
Supplier checks must verify these improvements.

Figure 2. What 1–2 star reviews of brick flowers mention



30% **Missing pieces**
across 111 brands

17% **Falls apart or droops**
across 96 brands

12% **Too hard, needed help**
across 87 brands

3 MY PLAN

SELL
Brick flowers
high review activity beyond LEGO

AND FIX WHAT BUYERS COMPLAIN ABOUT

1. **Checked counts, numbered bags**
missing pieces, 30%

2. **Sturdy stems**
falls apart, 17%

3. **Clear instructions**
too hard, 12%

HOW

Better quality and easier to build

Review activity suggests an opportunity.
Next: check suppliers and test-sell.

Amazon 2026: Is It Still True Today?

The reviews stop in 2023. A 2026 snapshot goes through the same pipeline and confirms the answer.



1 SAME PIPELINE

Keepa API · 1 h 27 min to fetch

2026 data: 724 flower toys on Amazon today, in the **same kinds I found in 2023**

3 LISTS — **335** top sellers today + **308** new since 2024 + **81** top products of 2023

STEP	2023 FIND	2026 VERIFY	
1 Ingest	Two files, read by Python + DuckDB	**Keepa API, fetched once**	NEW
2 Load	Meltano to BigQuery	Same, with 2 small tables	SAME
3 Store	Raw layer, never edited	Same raw layer	SAME
4 Transform	Four layers, star schema, flower rules	**Same rules, 1 new fact table**	ADDED
5 Test	dbt tests, then Great Expectations	**Same kinds, for the new tables**	ADDED
6 Dagster	Stage 1 of the run: find	**Stage 2, after the 2023 answer**	ADDED

STAR SCHEMA
one new fact table

fact_reviews
16.26M rows

same
22 kinds

NEW
fact_keepa_snapshot
724 rows

2 IS IT STILL TRUE?

Brick flowers from other brands, top 50 products

1 Is the interest still growing?

new ratings, Jan to 7 Oct: 2025 against 2026

2,124 → **6,255**

Yes. Almost three times as many new ratings.

2 Is the quality better?

average star rating, a year ago against now

4.5 → **4.5** **LEGO: 4.9**

No. The quality gap from 2023 is still open.

More interest, same quality problem

So the 2023 answer still holds:
sell brick flowers, and make them better.

AMAZON 2023 DATA · FIND | 2026 DATA · VERIFY

Dagster: One Run, Two Stages

First find on the 2023 data, then verify on the 2026 data. The same run can serve another category.



1 ONE RUN, TWO STAGES in the order I did the work

dagster

FIND Amazon 2023

1. Check the raw tables
2. Label the flower toys
3. Load the labels
4. dbt: build and test
5. Great Expectations
6. Reconcile the decision metrics

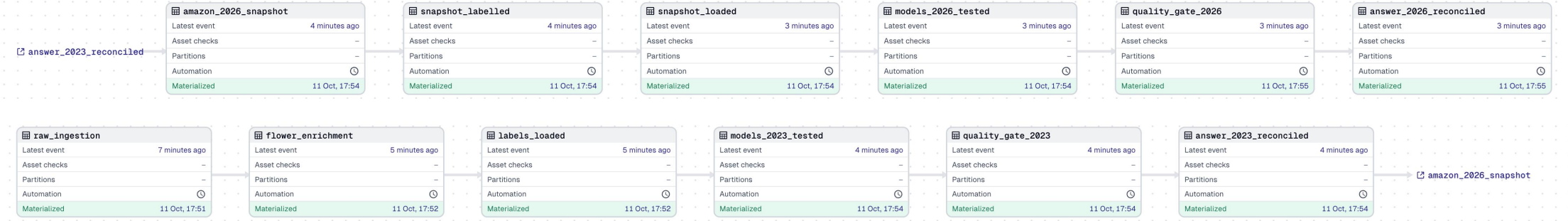
2023 checks pass before 2026 reads the saved snapshot. API collection was performed separately.

VERIFY Amazon 2026

7. Read the saved snapshot
8. Label the products
9. Load the snapshot
10. dbt: build and test
11. Great Expectations
12. Reconcile the decision metrics

2 THE SAME RUN IN DAGSTER screenshots

12 of 12 steps passed



Reuse the orchestration structure with category-specific inputs, labels and validation rules.
Another category still needs its own data, label checks and business-rule settings.